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Question 1.

Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Answer:

To create 4 equal subnets from a /24 network, borrow 2 bits.

Original Network: 192.168.10.0/24

New Subnet Mask: /26

Subnet Mask: 255.255.255.192

<i>Subnet</i>	<i>Network Address</i>	<i>Broadcast Address</i>	<i>Subnet Mask</i>
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<i>Subnet 1</i>	<i>192.168.10.0</i>	<i>192.168.10.63</i>	<i>255.255.255.192</i>
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<i>Subnet 2</i>	<i>192.168.10.64</i>	<i>192.168.10.127</i>	<i>255.255.255.192</i>
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<i>Subnet 3</i>	<i>192.168.10.128</i>	<i>192.168.10.191</i>	<i>255.255.255.192</i>
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<i>Subnet 4</i>	<i>192.168.10.192</i>	<i>192.168.10.255</i>	<i>255.255.255.192</i>
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Question 2.

For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.

Answer:

Using Subnet 1 (192.168.10.0/26)

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Usable Range: 192.168.10.1 – 192.168.10.62

Device IP Address

Router 192.168.10.1

Laptop 1 192.168.10.10

Laptop 2 192.168.10.11

Printer 192.168.10.20

Mobile 192.168.10.30

Note:

Network address (.0) is not assigned.

Broadcast address (.63) is not assigned.

Question 3.

Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Answer:

Router

192.168.10.1

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| | | |

Laptop1 Laptop2 Printer Mobile

192.168.10.10

192.168.10.11

192.168.10.20

192.168.10.30

Subnet 1: 192.168.10.0/26

Usable Range:

192.168.10.1 - 192.168.10.62

Other Subnets

Subnet 2: 192.168.10.64 - 192.168.10.127

Subnet 3: 192.168.10.128 - 192.168.10.191

Subnet 4: 192.168.10.192 - 192.168.10.255

Question 4.

Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Answer:

Step 1: Start with the Given Network

Network: 192.168.10.0/24

A /24 network has 256 total IP addresses.

Step 2: Divide into 4 Equal Subnets

To create 4 subnets:

$$2^2 = 4$$

Borrow 2 bits from the host portion.

$$/24 + 2 = /26$$

New subnet mask:

255.255.255.192

Step 3: Calculate Hosts Per Subnet

Remaining host bits:

$$32 - 26 = 6 \text{ bits}$$

Total addresses:

$$2^6 = 64 \text{ addresses}$$

Usable hosts:

$$64 - 2 = 62 \text{ hosts}$$

(One address is for the network and one for the broadcast.)

Step 4: Find Subnet Ranges

Block size:

$$256 - 192 = 64$$

Subnets increase by 64:

0

64

128

192

So the four subnets are:

192.168.10.0/26

192.168.10.64/26

192.168.10.128/26

192.168.10.192/26

Step 5: Assign Device IP Addresses

Choose addresses from the usable range:

Router = 192.168.10.1

Laptop1 = 192.168.10.10

Laptop2 = 192.168.10.11

Printer = 192.168.10.20

Mobile = 192.168.10.30

Conclusion:

New subnet mask = 255.255.255.192 (/26)

Number of subnets = 4

Usable hosts per subnet = 62

*Device IPs must be chosen from the usable host range,
not the network or broadcast address.*